# COM 5506

*L0. Introduction*

# Social Network Analysis for Communication

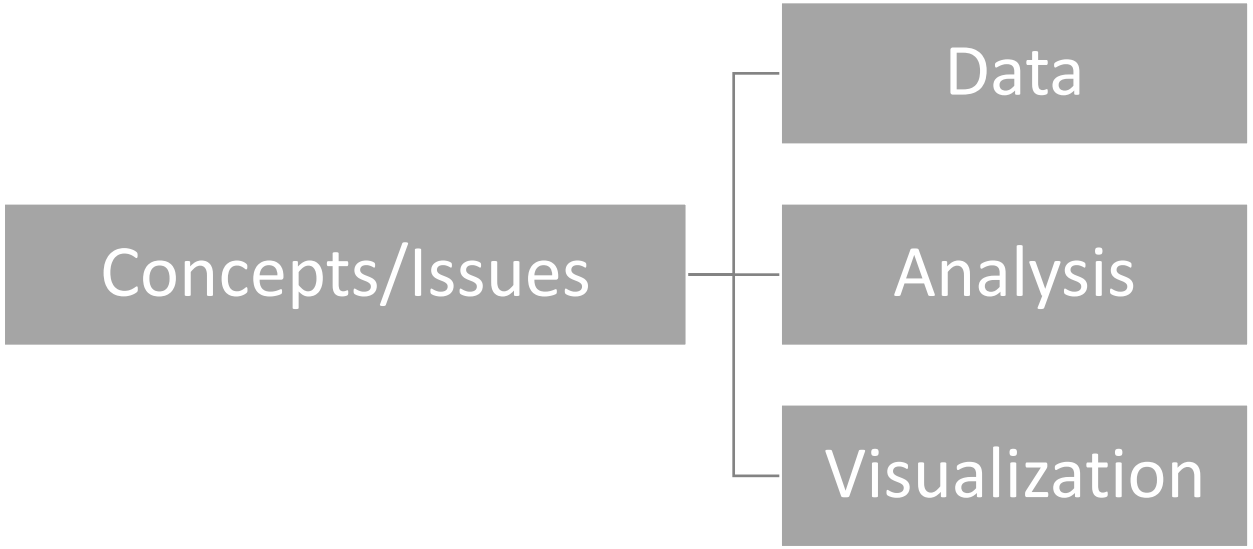
SEM B 2025/26

Fei Shen, City University of Hong Kong

# About this course & the stream

COM 5506

| | Stream Compulsory courses | |
|---|---|---|
| COM 5506 | Social Network Analysis for Communication | Network data understanding |
| COM 5507 | Social Media Data Acquisition and Processing | Web data scraping & text analysis |
| COM 5508 | Media Data Analytics | Data mining/machine learning |



# Syllabus highlights

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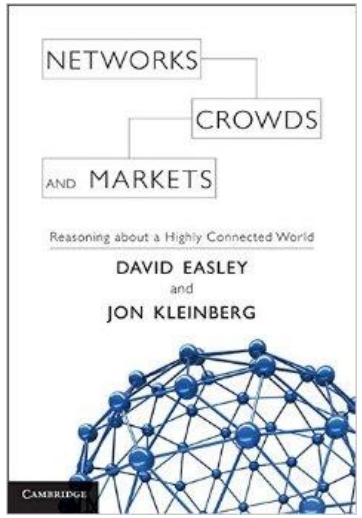
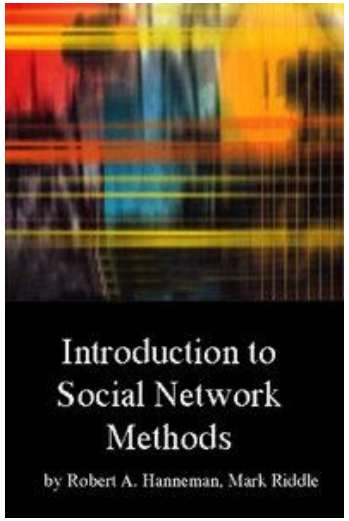
- Course textbooks

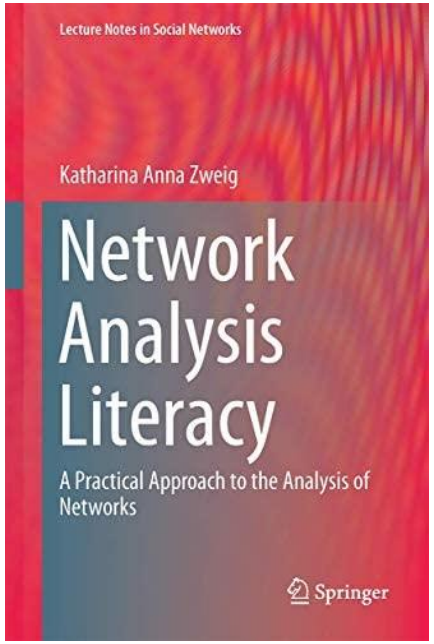



EK




HR

TK




- All textbooks are available online for free.
- Additional reading will be assigned

# Evaluation

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- Quiz – 20 points  (closed book, closed notes)
- Test – 30 points  (closed book, closed notes)
- In-class exercises – 10 points (a combination of data analysis, programming, and writing)
- Research project – group project (30 points) and presentation (10 points)
- Other issues
  - TA: Wei Yuhan, PhD student
  - Email:  yuhanwei5-c@my.cityu.edu.hk
  - Late assignments
  - Academic Honesty
  - General Politeness

# AI policy

- For this course, AI use is limited to a few scenarios.

- You can use AI to:
- 1. help you learn or solve technical issues such as coding
- 2. help you to proofread your assignment or project
- 3. help you review or learn knowledge related to this course

- You CANNOT use AI to:
- 1. Write your final project
- 2. Complete your homework for you

# Research paper

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- Each group consists 5-7 students. (PhD students should work independently without a group.)
- The project should address a theoretical or practical question based on empirical analysis of real world data collected by yourselves (preferably), from an open source or mixed of the two.
- Please send an email message to your TA by week 7 to report the following: the name and student ID of each member of the group.
- The paper is due April 22th. Students in the same group will receive the same grade for the project.

# Schedule

| Week | Date | Topic | Readings/Dues |
|---|---|---|---|
| 1 | 1/14 | Lecture: Social network and Graph<br>Lab: Network data structure | HR Chap 1-3 & Chap 5, EK Chap 2, TK Chap 1 |
| 2 | 1/21 | Lecture: Node centrality and power<br>Lab: Python introduction | HR Chap 7-8, Chap 10, EK Chap 4, TK Chap 3 |
| 3 | 1/28 | Lecture: Random graph networks<br>Lab: Python | PNAS-2002-Newman-2566-72 |
| 4 | 2/4 | Lecture: Small-world networks<br>Lab: Python | EK Chap 20, TK Chap 2 |
| 5 | 2/11 | Lecture: Scale-free networks<br>Lab: Python | EK Chap 18, TK Chap 4 |
| | 2/18 | **Lunar New Year Holiday Observed** | No class |
| 6 | 2/25 | Quiz (Lecture 1 to Lecture 4)<br>Lab: Data analysis practice | EK Chap 3, TK Chap 6 |
| 7 | 3/4 | **Reading week** | No class<br>Deadline for grouping – send to TA |
| 8 | 3/11 | Lecture: Strength of weak ties<br>Lab: Textual analysis example | EK Chap 16, TK Chap 5 |
| 9 | 3/18 | Lecture: Community detection<br>Lab: Gephi introduction<br>Workshop 1: project proposal consultation | EK Chap 19, TK Chap 6<br>Bring your topics and ideas to class |
| 10 | 3/25 | Lecture: Network data hypothesis testing<br>Lab: UCINET introduction | |
| 11 | 4/1 | Workshop 2: Group project progress consultation | Bring your data set to class |
| 12 | 4/8 | Final test (Lecture 1 to Lecture 8)<br>Workshop 3: Group project finding consultation | Bring your preliminary findings to class |
| 13 | 4/15 | *Final Project Presentation Day* | Complete your project draft |
| | | | |
| 14 | 4/22 | | **Research paper due** |

Gemini

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Gephi

UCINET Software

python™

ANACONDA
Powered by Continuum Analytics

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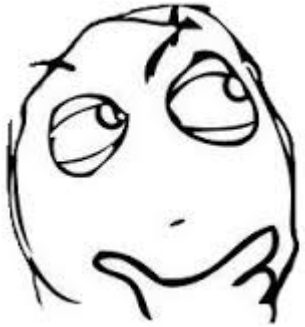
# Questions?

# Shall I drop this course?



# Self-evaluation

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1. Did you take any classes or get trained in programming – whatever language, either software or hardware? yes = 1
2. Did you take any university level statistics or math courses? yes = 1
3. Do you know the formula to calculate correlation coefficient? yes = 1
4. A fair coin is thrown in the air four times. If the coin lands with the head up on the first three tosses, what is the probability that the coin will land with the head up on the fourth toss? Know the correct answer = 1
5. What is the number of ways of choosing 2 items out of 5? Know the correct answer = 1
6. $\lim_{n\to\infty}\left(1+\frac{1}{n}\right)^n$ = ? Know the correct answer = 1
7. $\begin{bmatrix}1 & 2\\3 & 4\end{bmatrix}\cdot\begin{bmatrix}5 & 6\\7 & 8\end{bmatrix}$ =? Know the correct answer = 1
8. What is the result of these lines of codes?

```
x = 0
y = 0
def incr(x):
    y = x + 1
    return y
incr(5)
print x, y
```

## Friendly reminder:

**Score 7-8:** you are in excellent standing – follow my lecture, do the homework, read the reading, and you can pass the course easily

**Score 5-6:** you are in okay shape – you might need to spend extra time to work through this course, but if you study hard, you should be able to pass the course.

**Score 3-4:** you should reconsider your choice – You need to input a great amount of time to understand the course. You might want to use this course as your elective if not dropping it.

**Score 1-2:** …

# Class data collection now. Please fill out the questionnaire carefully.

The data will be anonymized and only used for our own learning.

# COM 5506 *L 1. Social network and Graph*

# Social Network Analysis for Communication

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# Social network is everywhere. Can you give an example of social network?

# Social media ego network mapping

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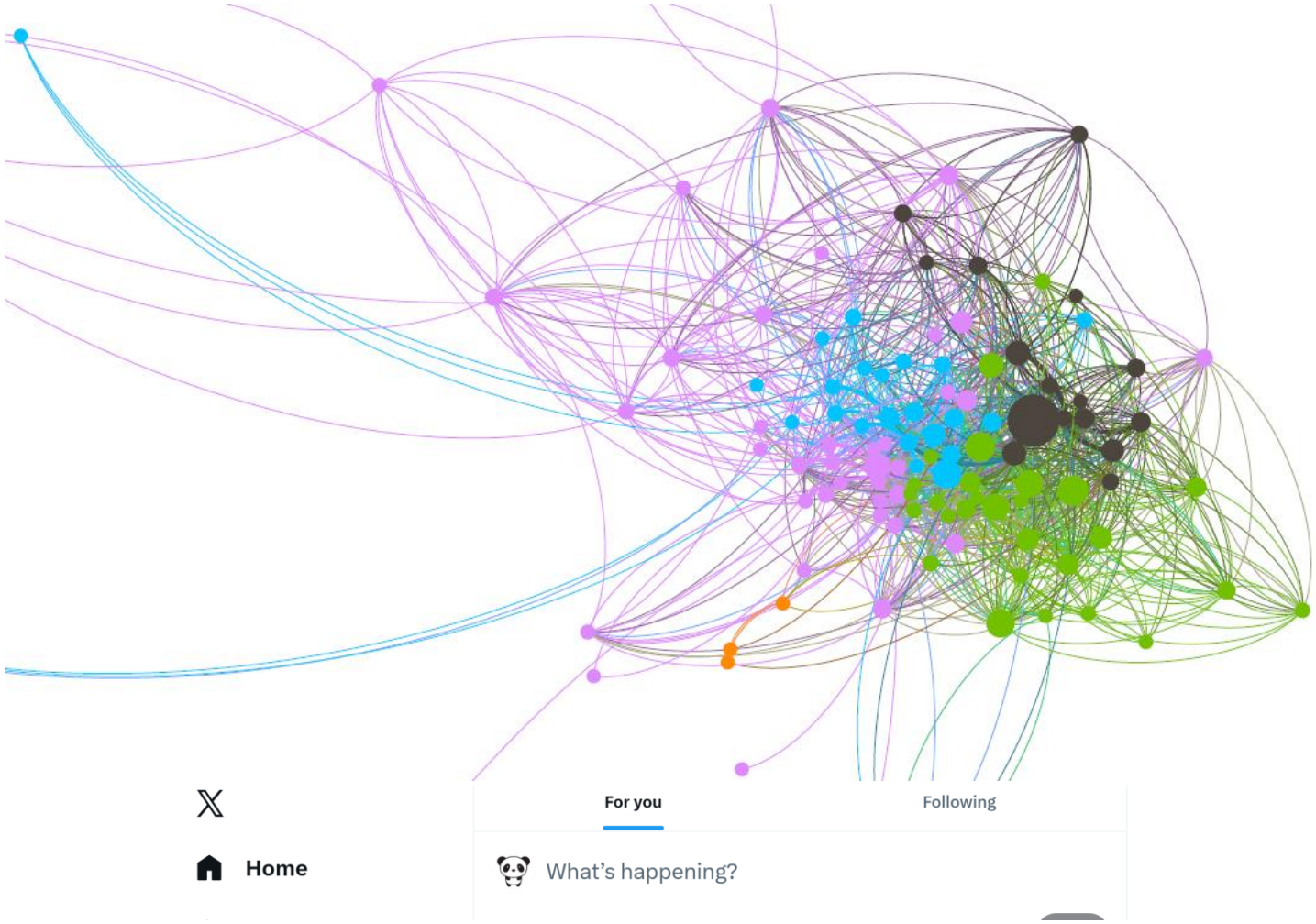


- This is my Twitter/X network
- Relatively compact
- Explicit network

# The Dream of the Red Chamber power network

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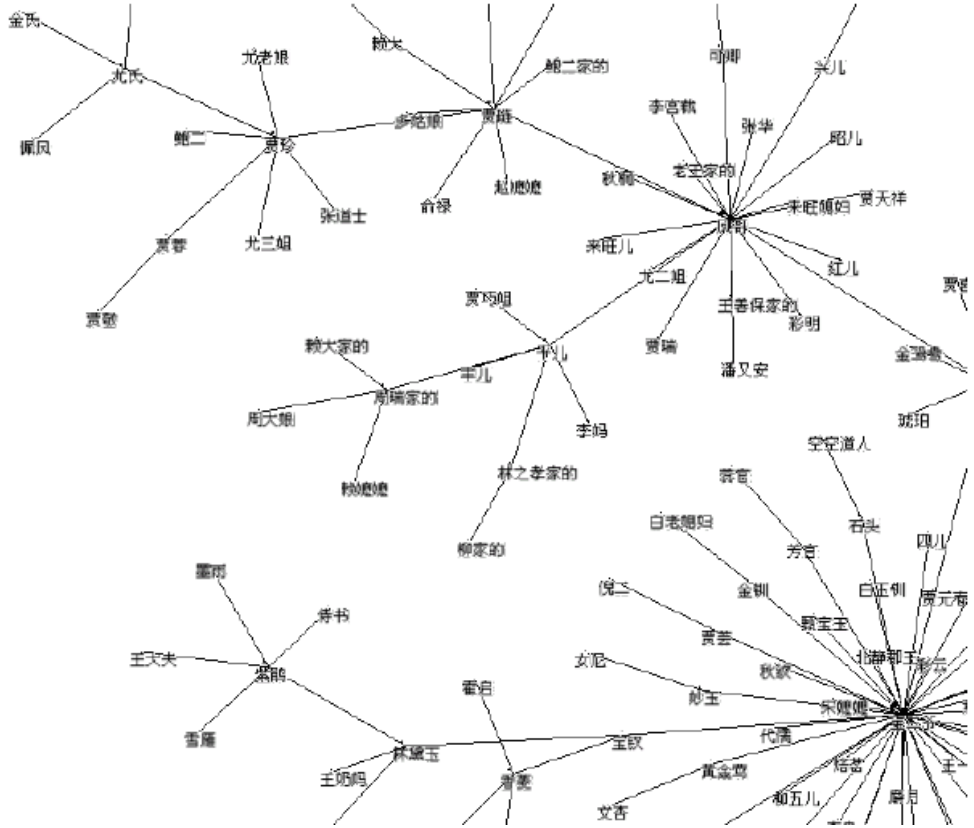


- Power relation network
- Implicit network

## 《红楼梦》中社会权势关系的提取及网络构建


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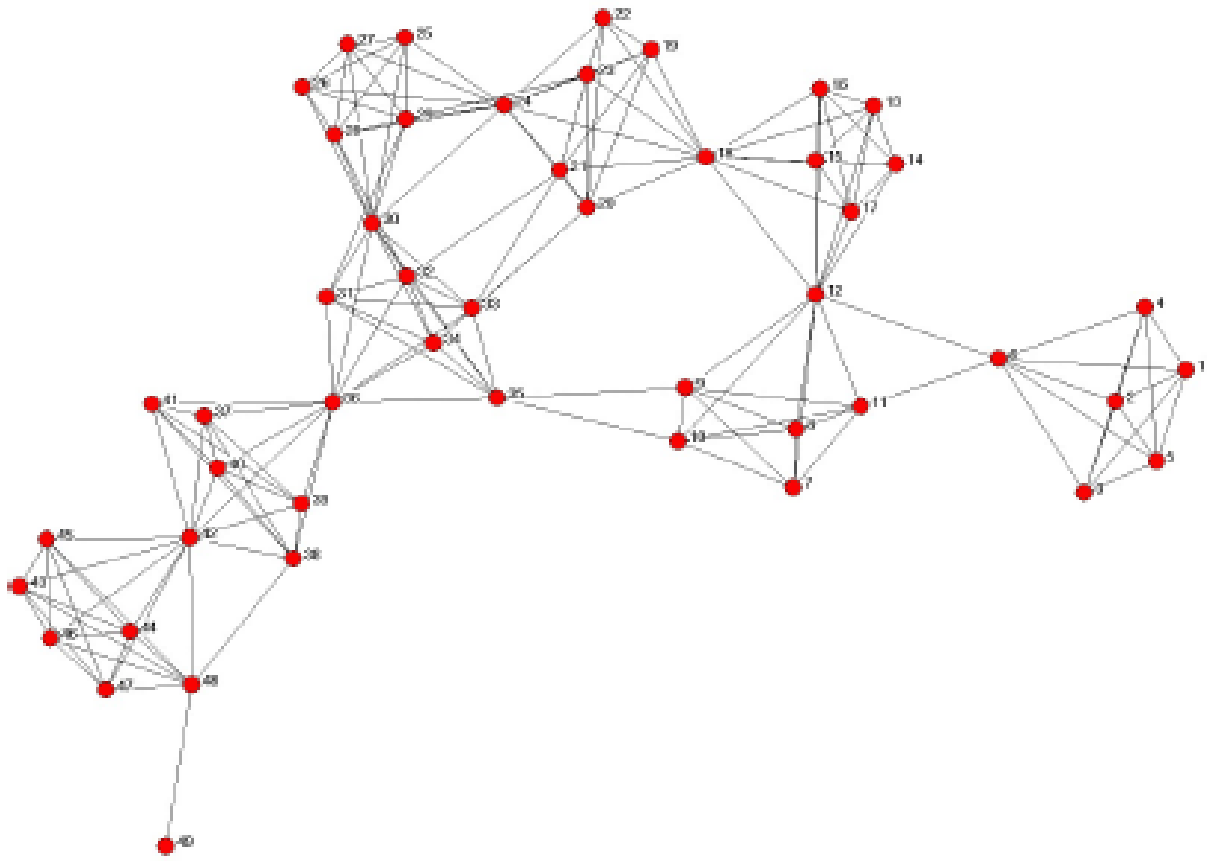
[2]计算语言学教育部重点实验室，北京，100871)

Email: {1100012154, 1300011764, aiwei, hujf}@pku.edu.cn



**摘 要：** 社会地位与权势的研究一直是社会语言学领域的一个热点话题。本文借助数据挖掘中的关系提取方案雪球算法（Snowball Algorithm），实现了红楼梦文本中候选的特征语言模式（pattern）和人物关系对之间的相互定位与赋权，对小说中频繁同现的人物对之间的社会等级关系进行挖掘，以此建立了能反映人物等级关系的有向加权人际关系网络。进一步应用最小树形图算法，生成了涵盖 192 个红楼梦主要人物的单向联通的树状社会关系图。通过这种方法生成的社会关系图不但能有效反映人际交往亲密度与社区影响力，同时还透视了人与人之间的社会等级差异。相较于单纯基于人际交往亲密程度的无向关系网络，能更加客观的表达出社会交往中人际关系网络的真实图景。

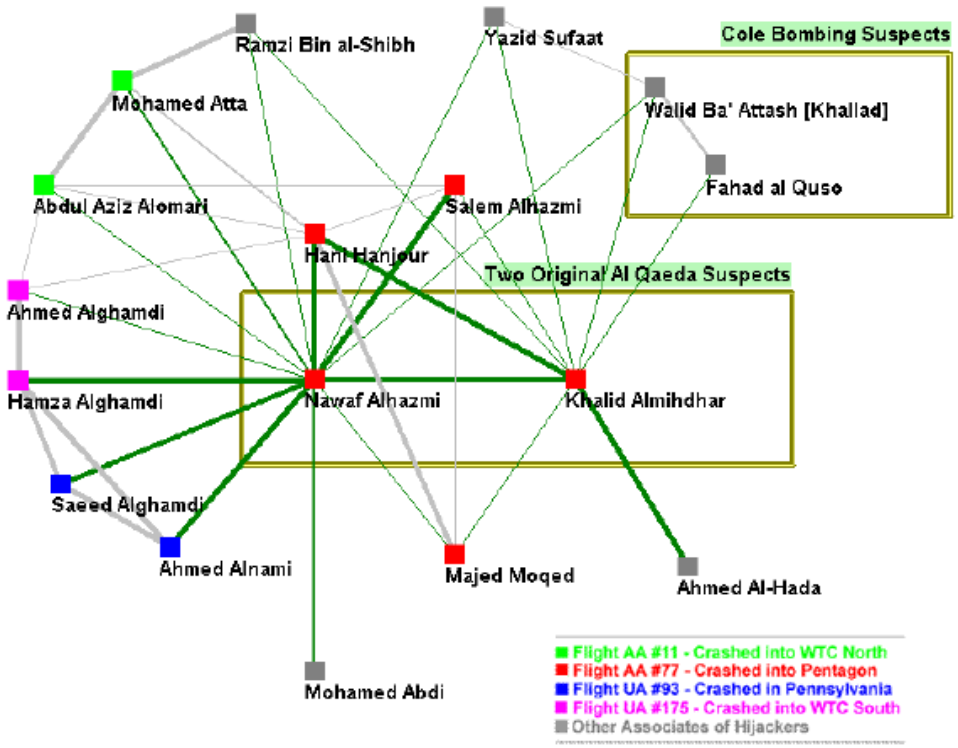
# Structure of terrorist network



- Secret network
- 6 or fewer members in each cell
- Isolated from the rest of the graph before action

Simulated based on Al Qaeda operations manual. Tsvetovat and Carley (2005). Journal of Social Structure.

# 911 Terrorists' network




9/11 terrorist network
[Krebs, 2002]

- Network visualization can offer us much information regarding the structure of the network and the organization of personnel

# Network analysis for organizational restructure

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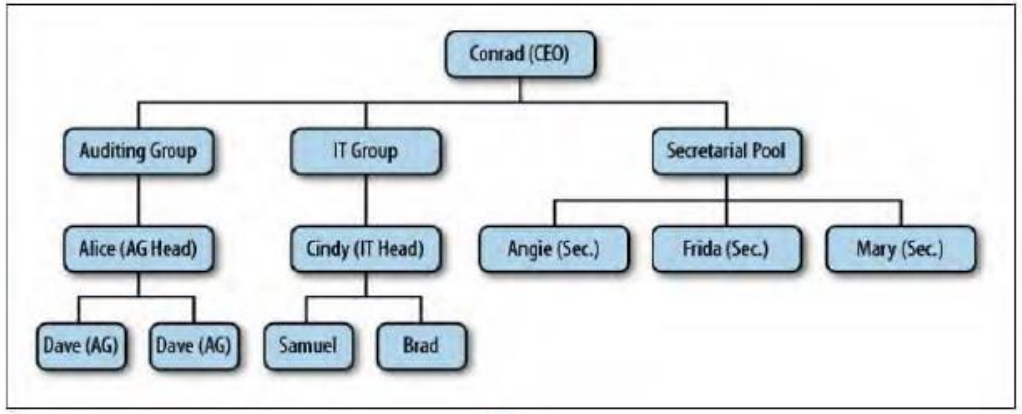
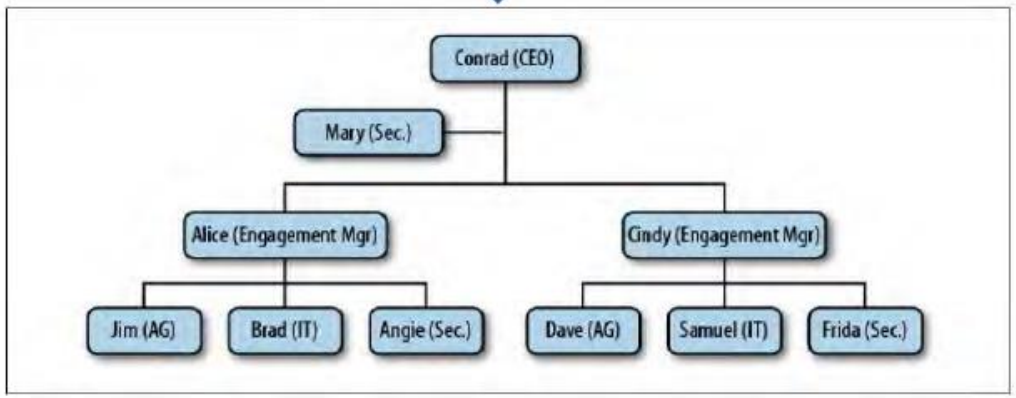
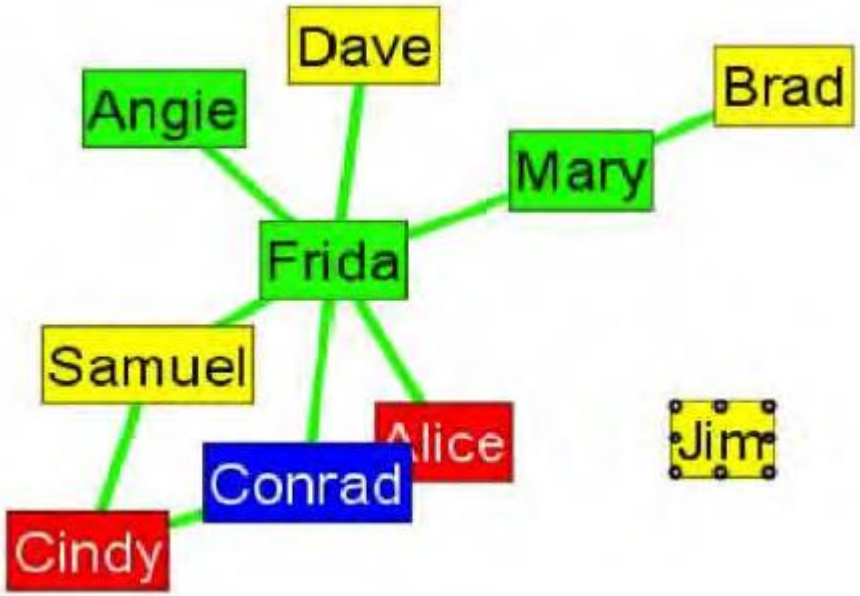
Before




Reorganization

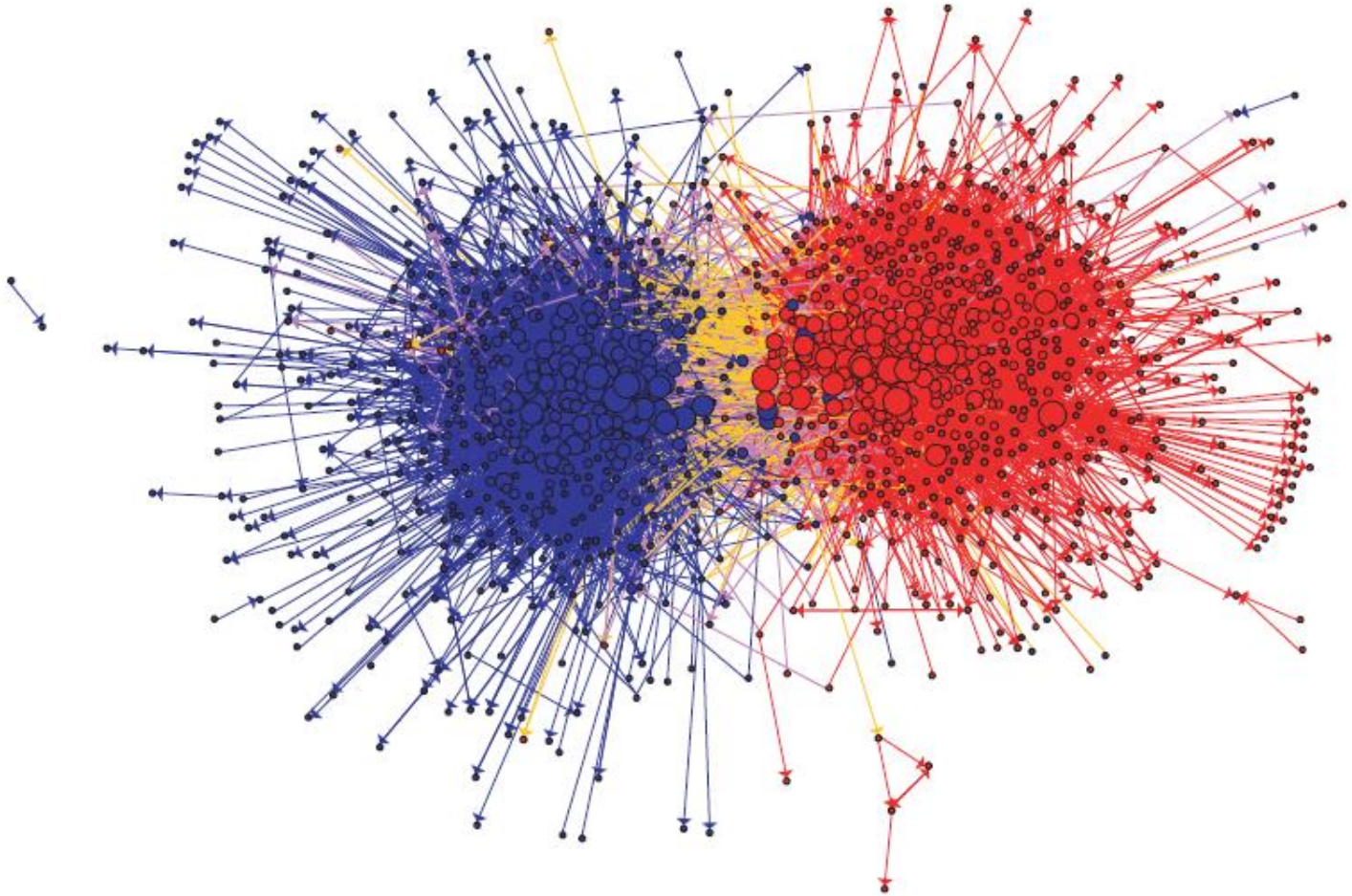
After




Source: TK (2011)

# US blogosphere mapping



- From person-to-person network to blog-to-blog network
- Political blogs of U.S. liberal and conservative bloggers
- A link exists if one blog cites another
- Homophily

# The 147 Companies that control everything

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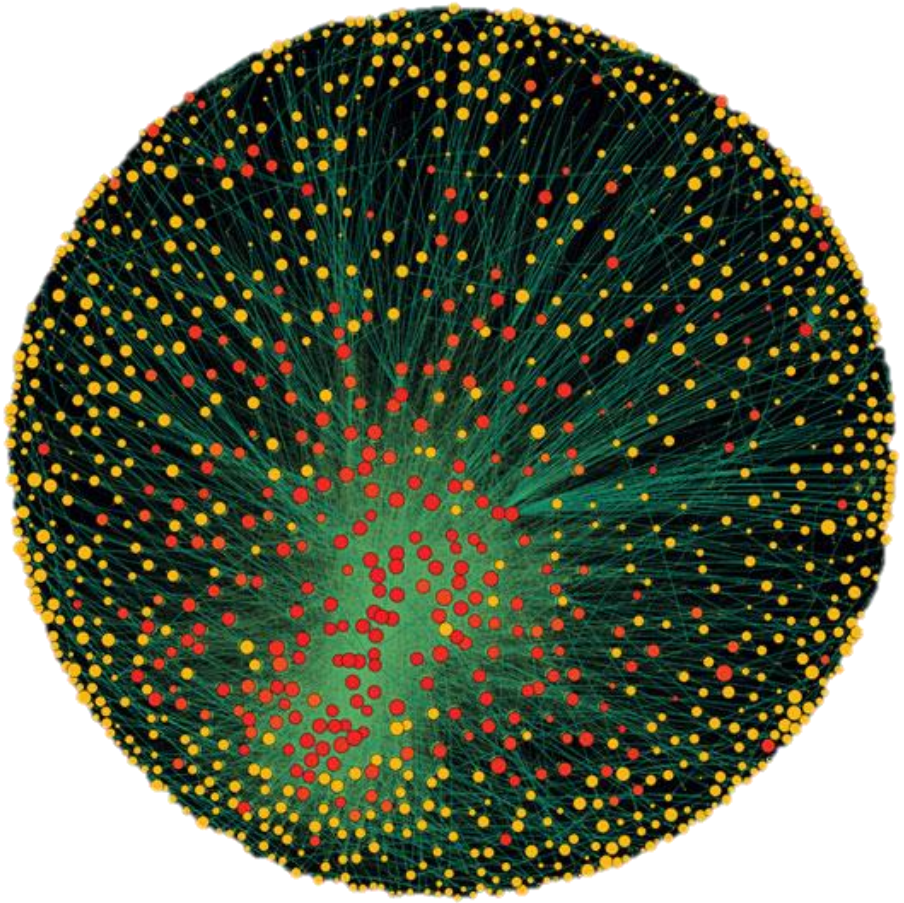
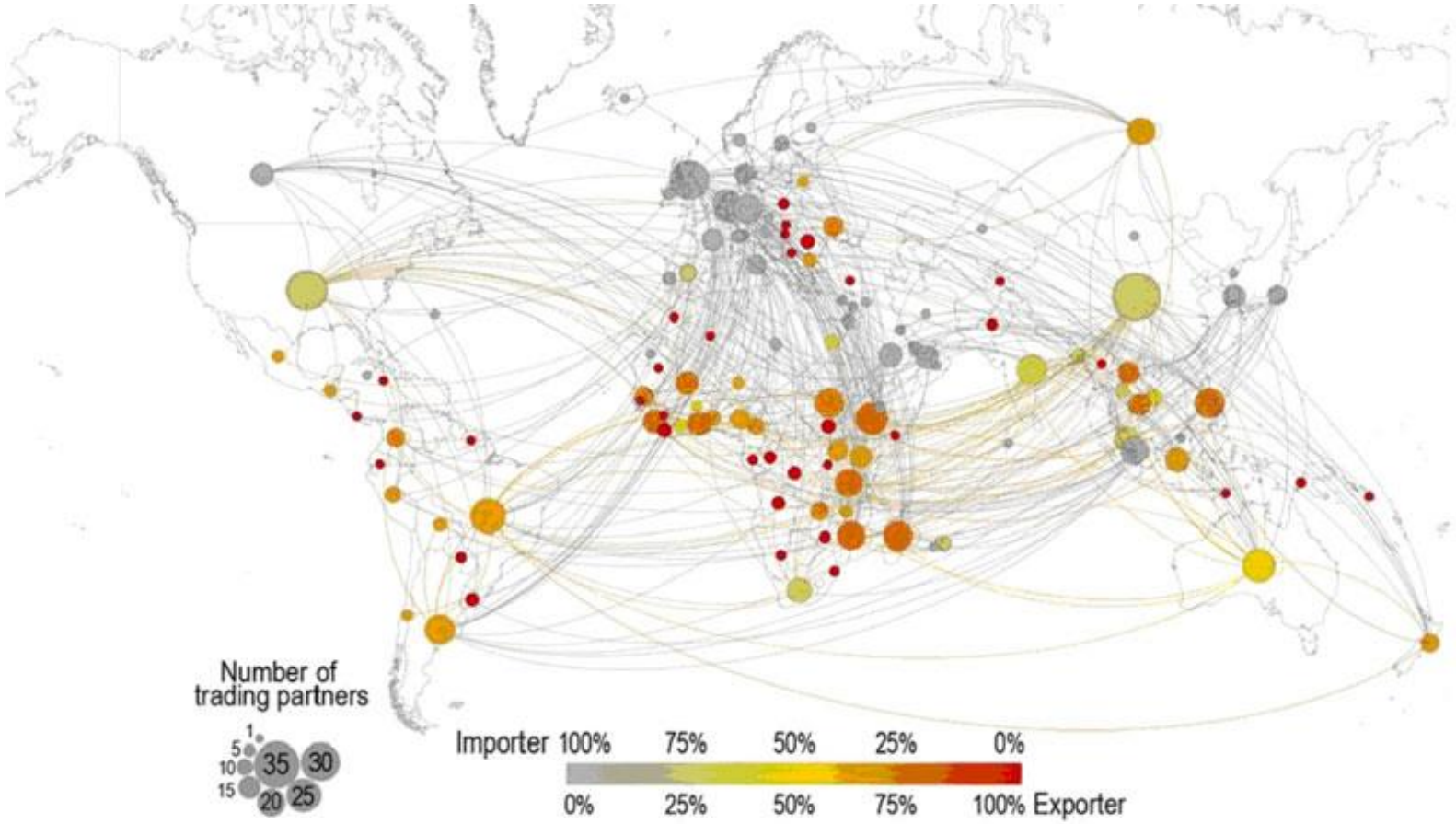


- Organizational network
- An analysis of the relationships between 43,000 transnational corporations has identified a relatively small group of companies, mainly banks, with disproportionate power over the global economy.
- share ownerships forms links
- The 1318 transnational corporations that form the core of the economy.
- Superconnected companies are red, very connected companies are yellow. The size of the dot represents revenue.

# Network of international trade




- Trading partner network
- Nation-states as nodes
- Goods and monetary flow as links

# Academic citation networks

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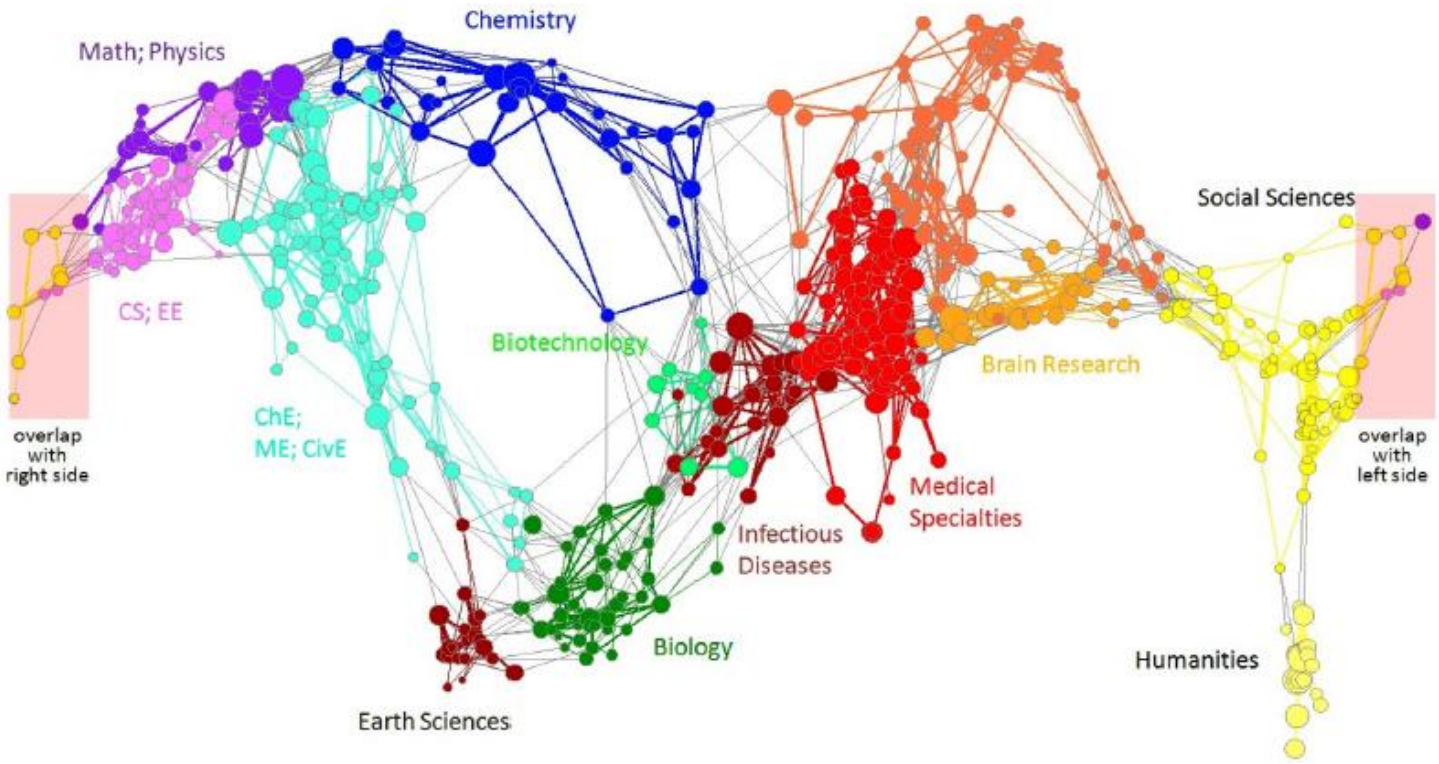
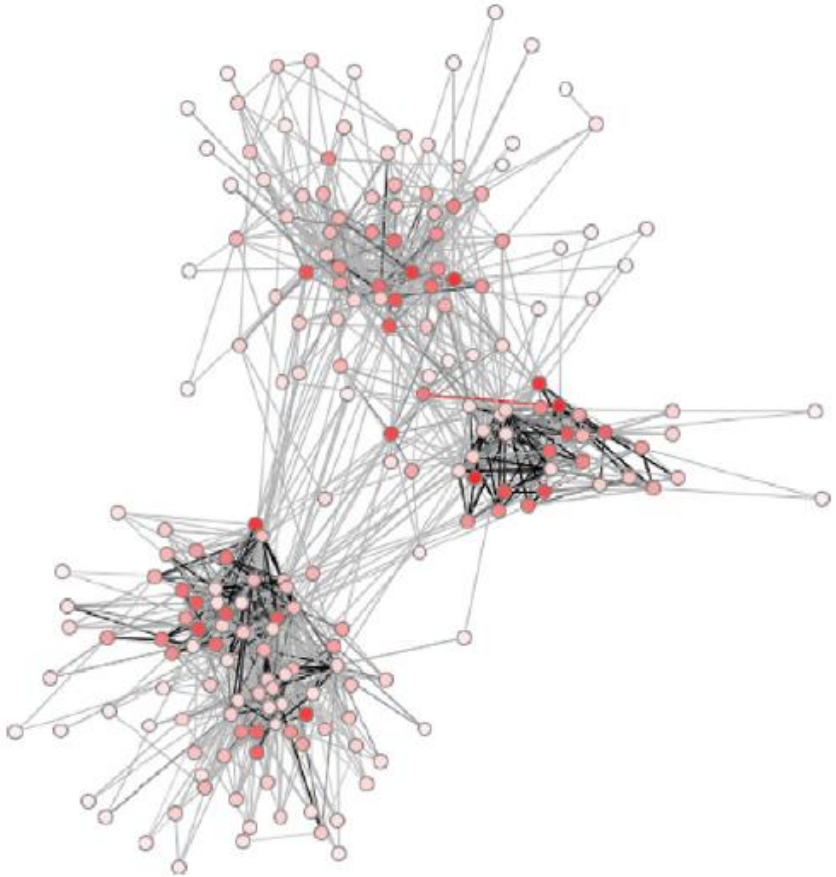



Citation networks and Maps of science
[Börner et al., 2012]

- Academic citation network shows us how different disciplines are related to each other and the structure of knowledge

# Music notes network



X.F. Liu et al. / Physica A 389 (2010) 126–132

- A sample network from Bach's violin solos.
- Darkness of coloring of nodes indicates relative degrees, and darkness of coloring of edges indicates relative weights.
- Help us to understand music better

# Networks are around us

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- Persons – kinship ties, friendship, facebook connections, intimate relationship
- Organizations – business relationship, trade flow
- Nations – international relationship
- Machines – web servers, routers
- Words – semantic relationship
- Location – highway network

# What do we learn here?

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- Communication and media cannot exist without networks. Everything is related to each other.
- It’s not about what you know; it’s about who you know and who you know knows who.
- Relations are at least as important as if not more important than attributes and characteristics.
- Informal relations are often more important than formal relations.

# Why popular now?

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- Network science
  - While the study of networks has a long history, with roots in graph theory and sociology, the modern chapter of network science emerged only during the first decade of the 21st century.
- Data availability and Computational power
  - In the past, we lacked the tools to map these networks. It was equally difficult to keep track of the huge amount of data behind them. The Internet revolution, offering effective and fast data sharing methods and cheap digital storage, fundamentally changed our ability to collect, assemble, share, and analyze data pertaining to real networks.
- Universality of network features
  - A key discovery of network science is that *the architecture of networks emerging in various domains of science, nature, and technology are similar to each other, a consequence of being governed by the same organizing principles. Consequently we can use a common set of mathematical tools to explore these systems*.

# How do we study networks?

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- Visualization
  - the dummy approach
  - just by looking
- Statistical/empirical: examine real network data to find patterns
  - bottom-up approach
  - e.g., find out the most important figure in a network
- Mathematical: graph theory and idealistic stats models
  - Top-down approach
  - e.g., if everyone has the equal chance of meeting each other in a network, under what circumstance will a big group form?

# How do we proceed?

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- What are the overall features of the network?
- Are nodes connected?
- How far apart are they?
- Which nodes are more important?
- Are their communities in the network?
- What are the process mechanism generating the network?
- Randomly generated networks
- Preferential attachment
- Small-world networks

# Understanding networks

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# Networks defined: terminology basics

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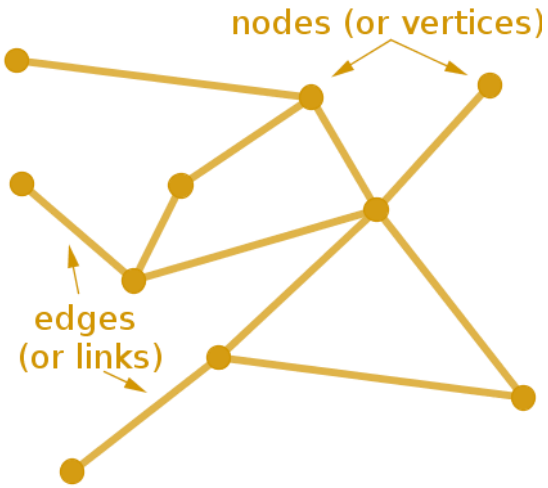
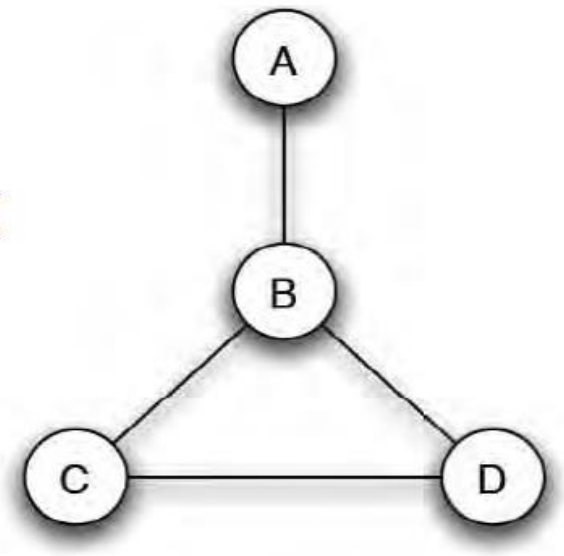
- Networks are sets of nodes connected by edges. Graphs are mathematical structures used to model pairwise relations between objects.




| Points | Lines | discinplines |
|---|---|---|
| vertices | edges, arcs | mathematics |
| sites | bonds | physics |
| actors | ties | sociology |
| nodes | links | computer science |
| nodes | edges | network science |

# Graph: mathematical representation of a network




(a) *A graph on 4 nodes.*

- The same set of nodes and edges can also be referred to as:
  - a network
  - a matrix
  - a web
  - a map
- Nodes adjacent to each others are called "*neighbors*."

# The origin of graph theory

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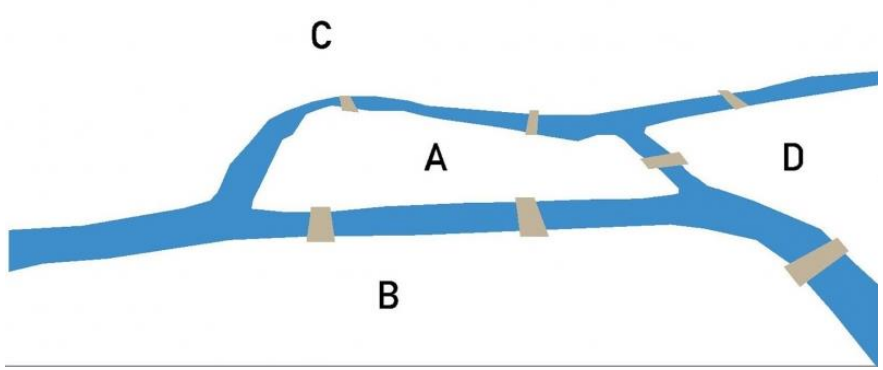
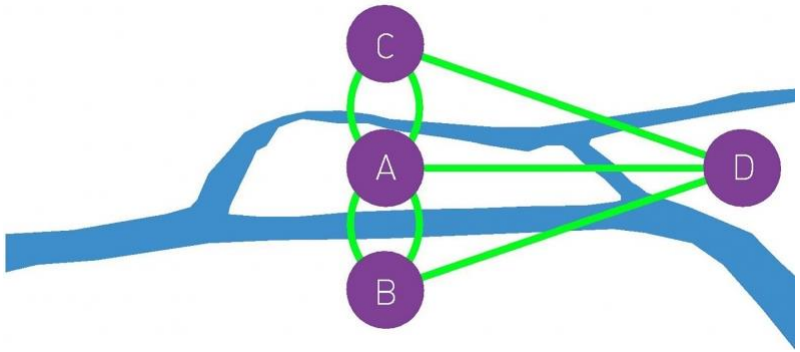
- Graph theory, the mathematical scaffold behind network science, can be traced back to 1735 in Königsberg, the capital of Eastern Prussia, a thriving merchant city of its time.
- Can one walk across all seven bridges and never cross the same one twice? Despite many attempts, no one could find such path. The problem remained unsolved until 1735, when Leonard Euler, offered a rigorous mathematical proof that such path does not exist .



# Map, Graph and Matrix of Arpanet

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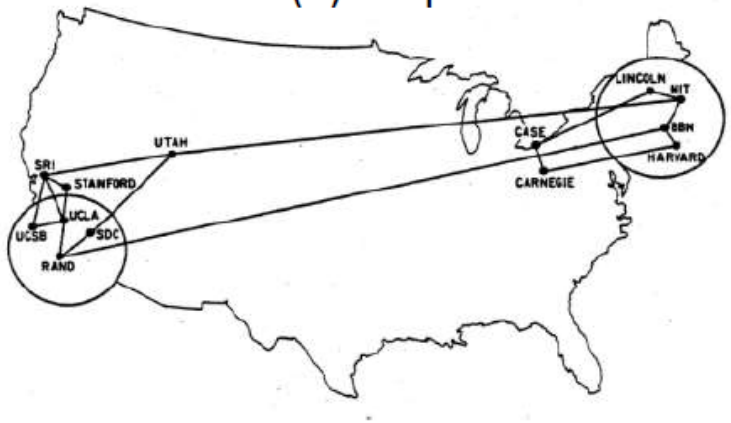
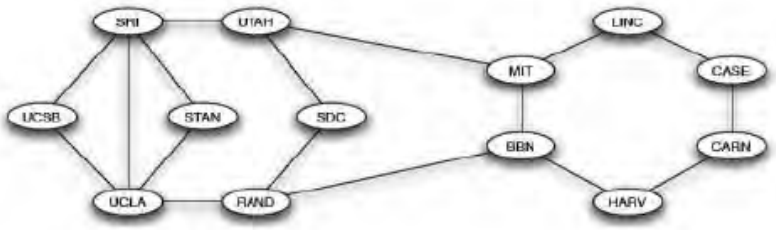
(a) Map




(b) Graph




(c) Matrix

| | UCSB | SRI | STAN | UCLA | ... |
|---|---|---|---|---|---|
| UCSB | | | | | ... |
| SRI | 1 | | | | ... |
| STAN | 0 | 1 | | | ... |
| UCLA | 1 | 1 | 1 | | ... |
| ... | ... | ... | ... | ... | ... |

Source: EK Fig 2.3

# Attributes of nodes and edges

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- Nodes can take on various types of characteristics or properties.
- For example:
  - Age
  - Sex
  - Occupation
  - Education

- Edges can have their own features as well.
- For example:
  - Presence (yes v.s. no)
  - Strength (weight)
  - Valence (positive or negative)
  - Direction (one-way or two-way)

# Edge attribute: directed or not

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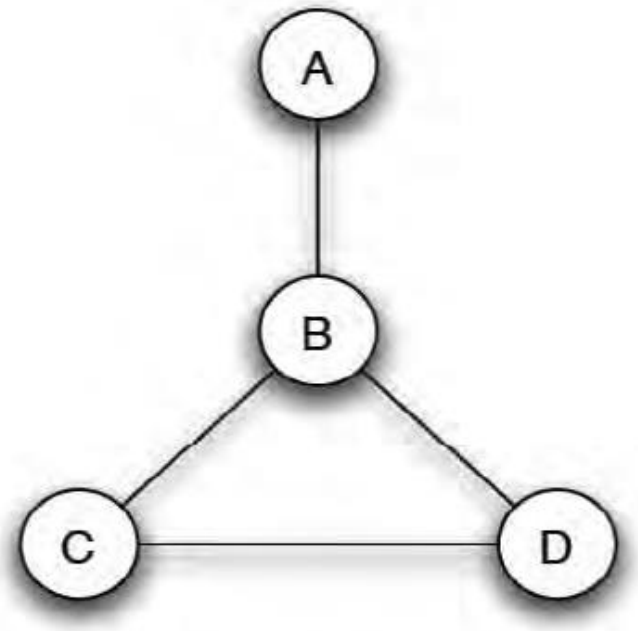


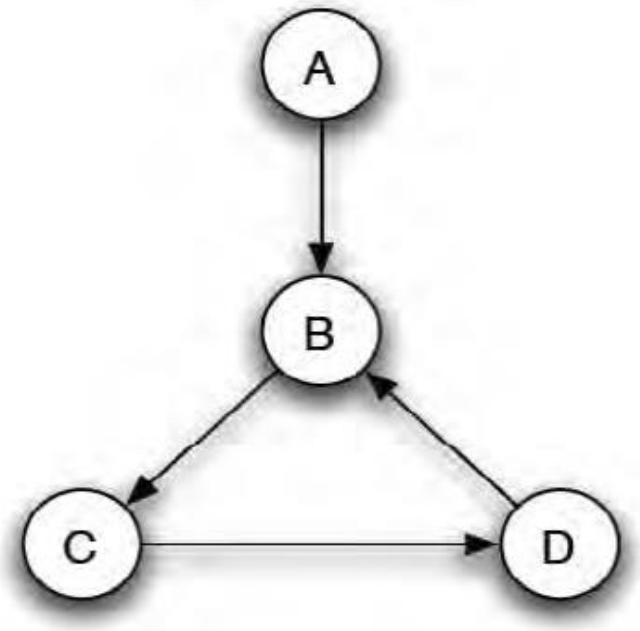

(a) *A graph on 4 nodes.*

facebook

twitter

新浪微博

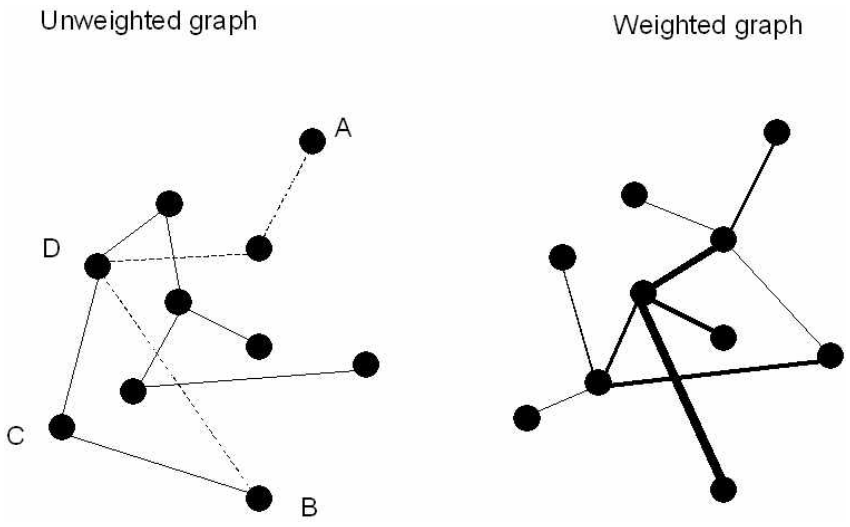
微信




(b) *A directed graph on 4 nodes.*

Figure 2.1: Two graphs: (a) an undirected graph, and (b) a directed graph.

# Edge attribute: Weight

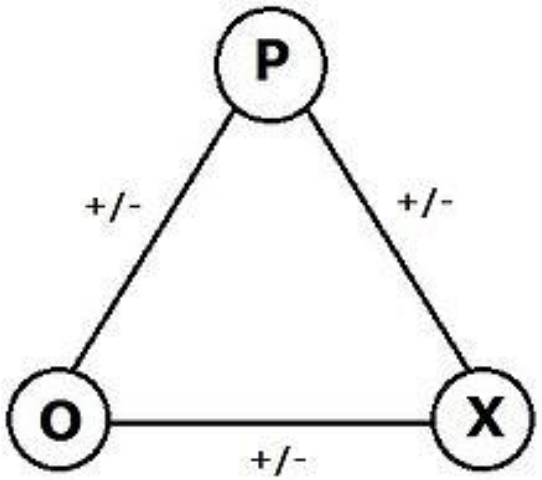



All edges share the same strength

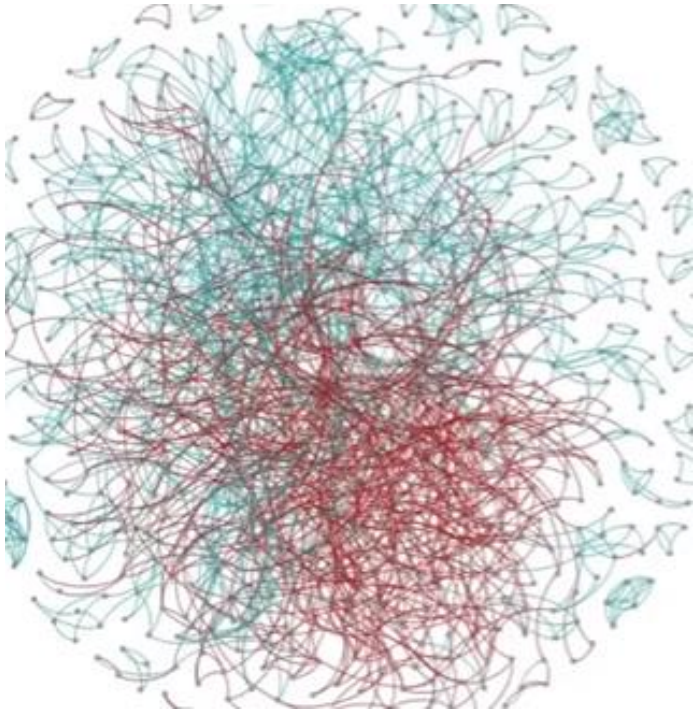
Edges vary in strength

# Edge attribute: Sign

Signed graph




Edges can take negative values



Sample of positive & negative ratings from Epinions network

# Direction, weight, and signs of graph

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| | Undirected | Directed |
|---|---|---|
| Unsigned | | |
| Unweighted | undirected-unweighted-unsigned graphs | directed-unweighted-unsigned graphs |
| Weighted | undirected-weighted-unsigned graphs | directed-weighted-unsigned graphs |
| Signed | | |
| Unweighted | undirected-unweighted-signed graphs | directed-unweighted-signed graphs |
| Weighted | undirected-weighted-signed graphs | directed-weighted-signed graphs |

# Single node property

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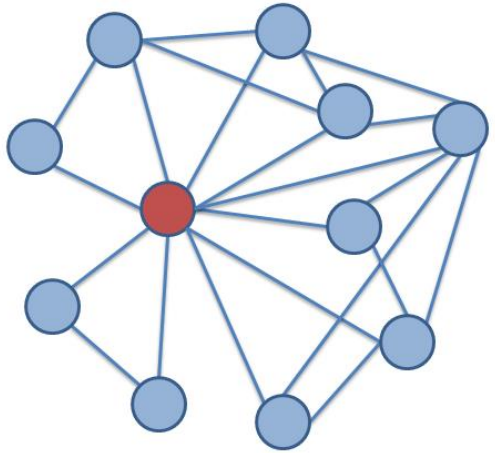
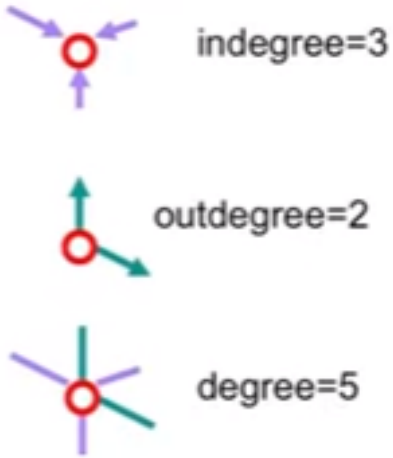
- *Degree*: how many edges a node has?

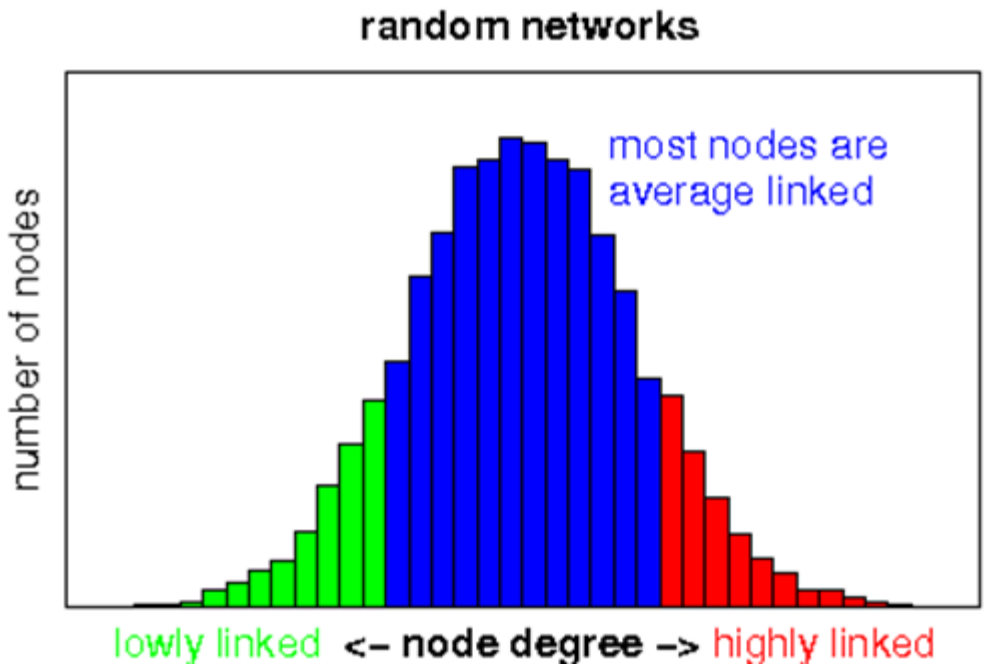


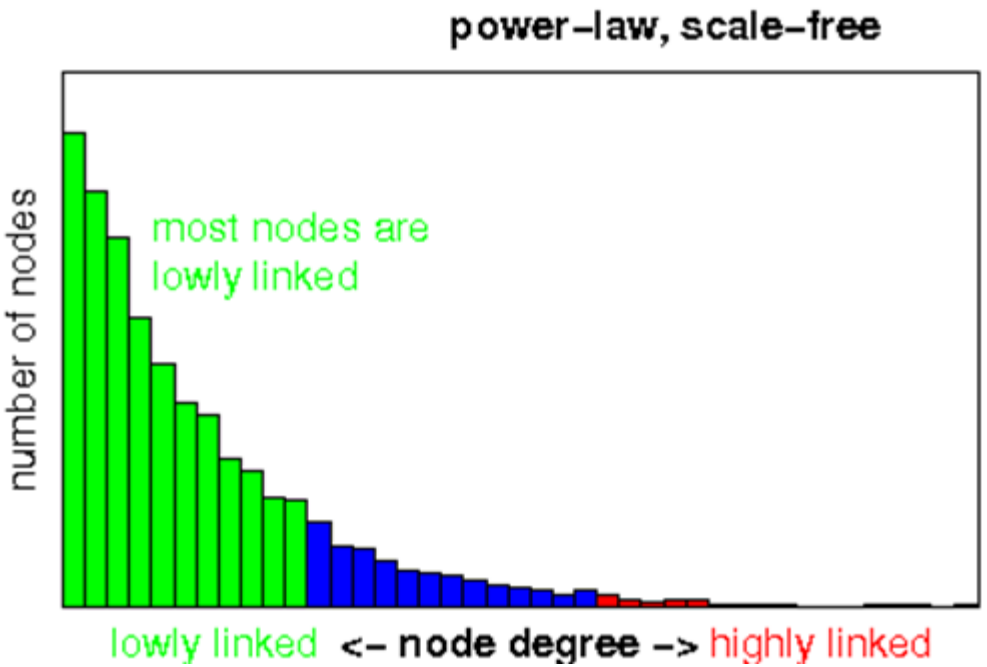
- *In degree*:
  - How many directed edges are incident on a node
- *Out degree*:
  - How many directed edges originate from a node



# Network property: degree distribution

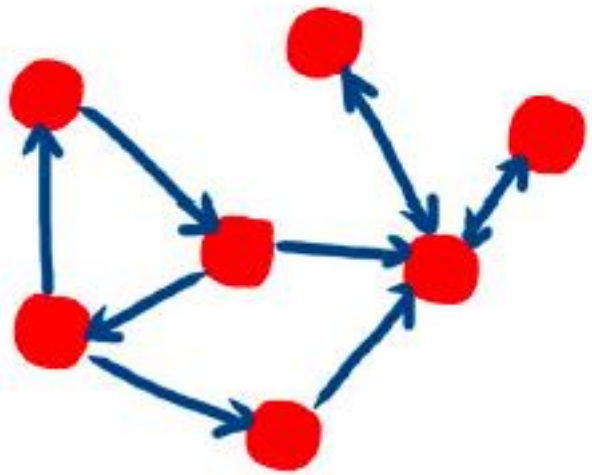
COM 5506

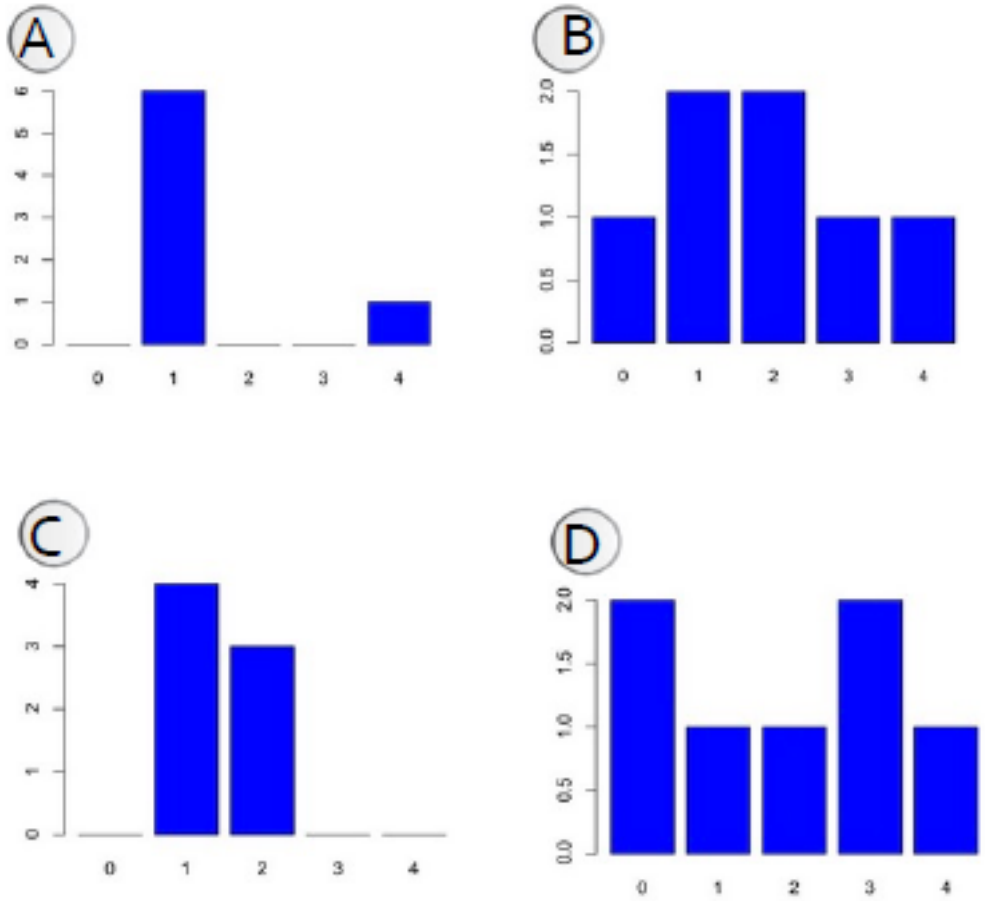


# Which histogram looks right?

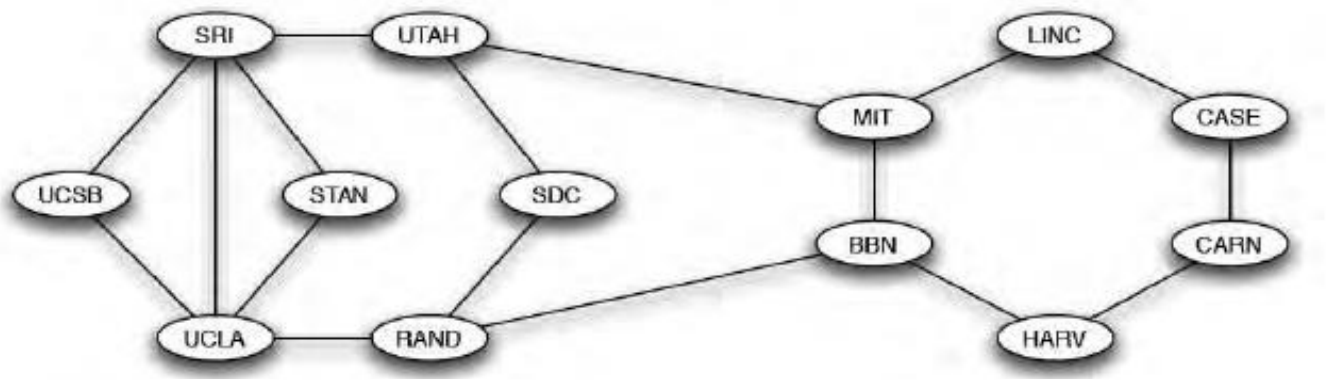
COM 5506





# Features between two nodes

- *Path*: a sequence of connected nodes
- *Distance*: the number of edges (steps) in the shortest path

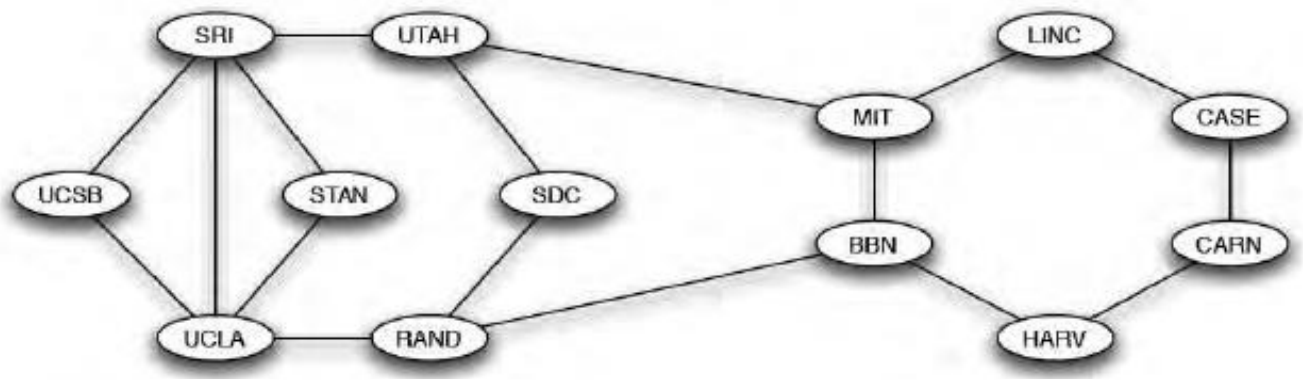



- How many paths are there between UCSB and SRI?
- What is the distance between CASE and RAND?

COM 5506

- *Diameter*: the maximum distance between any pair of nodes

- *Average path length*: the average distance over all pairs of nodes




- What is the diameter of the APARNET graph?

$$l_G = \frac{1}{n \cdot (n-1)} \cdot \sum_{i \neq j} d(v_i, v_j)$$

# Parts of a network: components

COM 5506

- A component is a subset of a graph
  - Every node in the subset has a path to every other
  - The subset is not part of some larger set
- Most online social networks have a giant component plus some isolated nodes

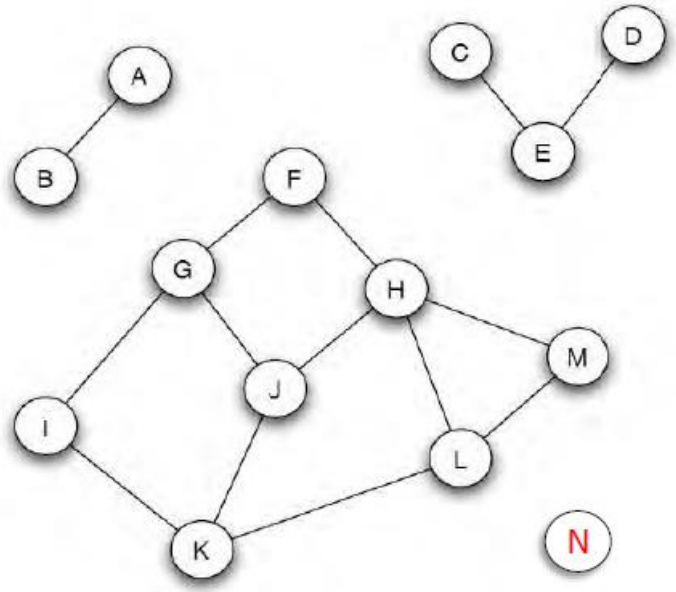



Figure 2.5: A graph with three connected components.

Is N a component?

# Highs school network

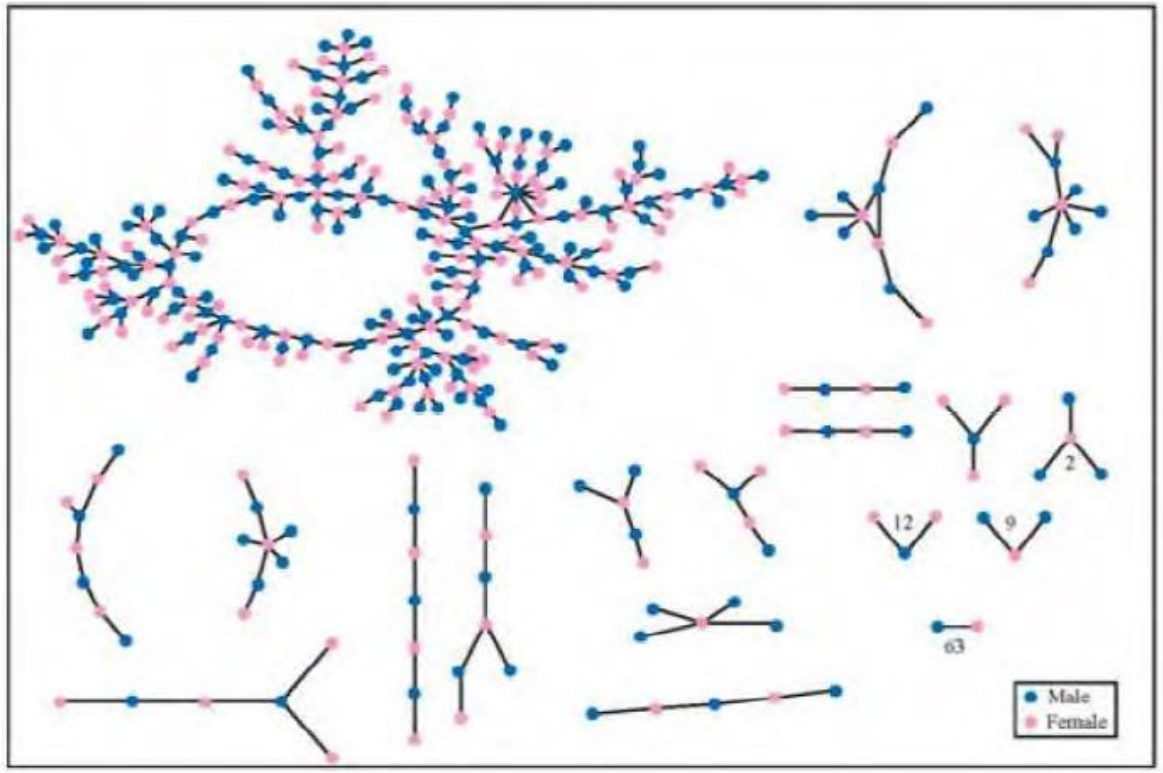



Figure 2.7: A network in which the nodes are students in a large American high school, and an edge joins two who had a romantic relationship at some point during the 18-month period in which the study was conducted [49].

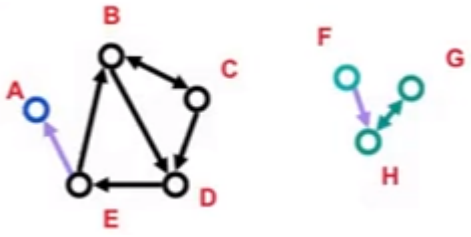
This chapter describes the structure of the adolescent romantic and sexual network in a population of over 800 adolescents residing in a midsized town in the midwestern United States. Precise images and measures of network structure are derived from reports of relationships that occurred over a period of 18 months between 1993 and 1995. The study offers a comparison of the structural characteristics of the observed network to simulated networks conditioned on the distribution of ties; the observed structure reveals networks characterized by longer contact chains and fewer cycles than expected. The chapter identifies the micromechanisms that generate networks with structural features similar to the observed network. Implications for disease transmission dynamics and social policy are explored.

Bearman, P. S., Moody, J., & Stovel, K. (2016). The Structure of Adolescent Romantic and Sexual Networks. *Handbook of Applied System Science*, 164.

# WCC and SCC

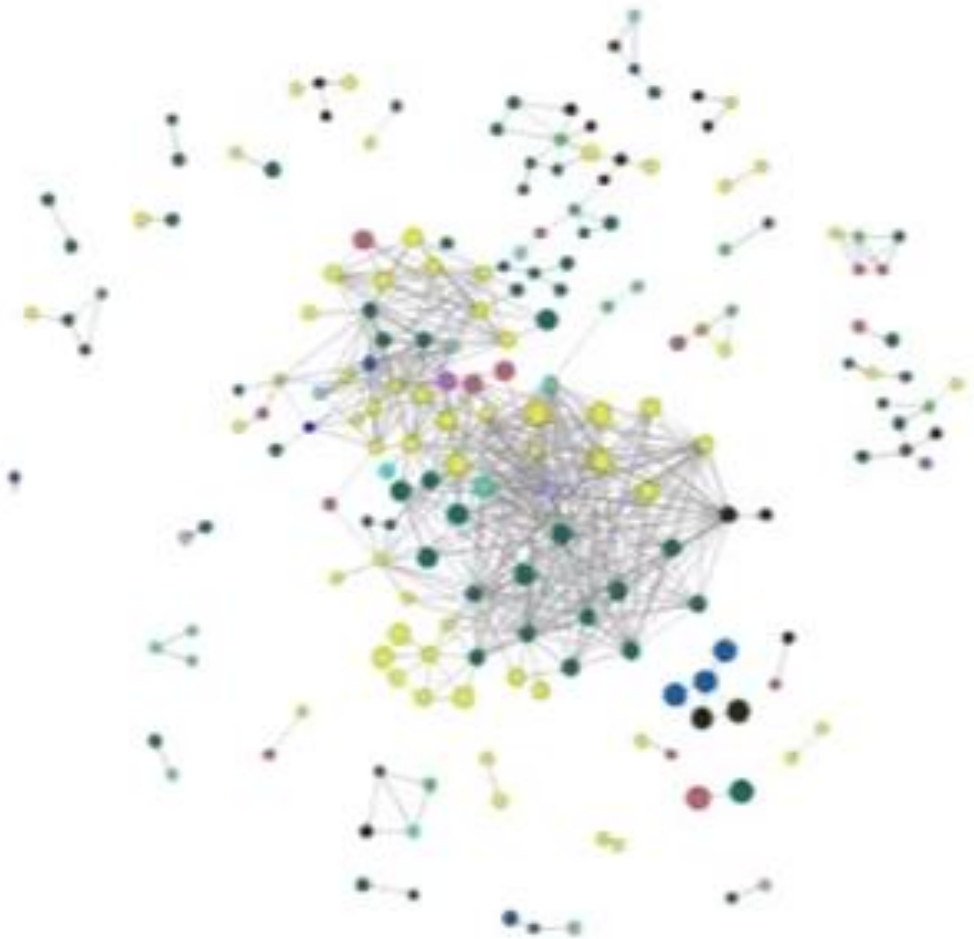
- Undirected graphs:
- By definition, every node has a path to every other node in the same component.
- In other words, every node can reach every other node in the same component.
- All components in any undirected graph are SCC.

- Directed graphs:
- *Weakly connected components* (WCC) – all nodes are connected but not all of them can reach every other node
- *Strongly connected components* (SCC) – all nodes are not only connected but also reachable by every other node.



# Giant component

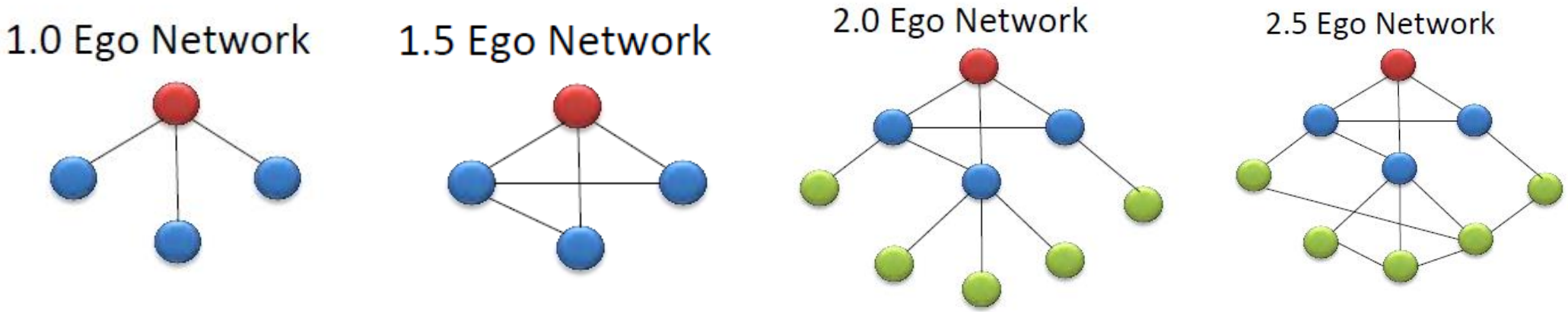
- If the largest component encompasses a significant fraction of the graph, it is called the giant component.
- A component is called small it it contains more than $n^{2/3}/2$ nodes and large if it contains more than $n^{2/3}$ nodes.
- The term giant component refers to the largest component.



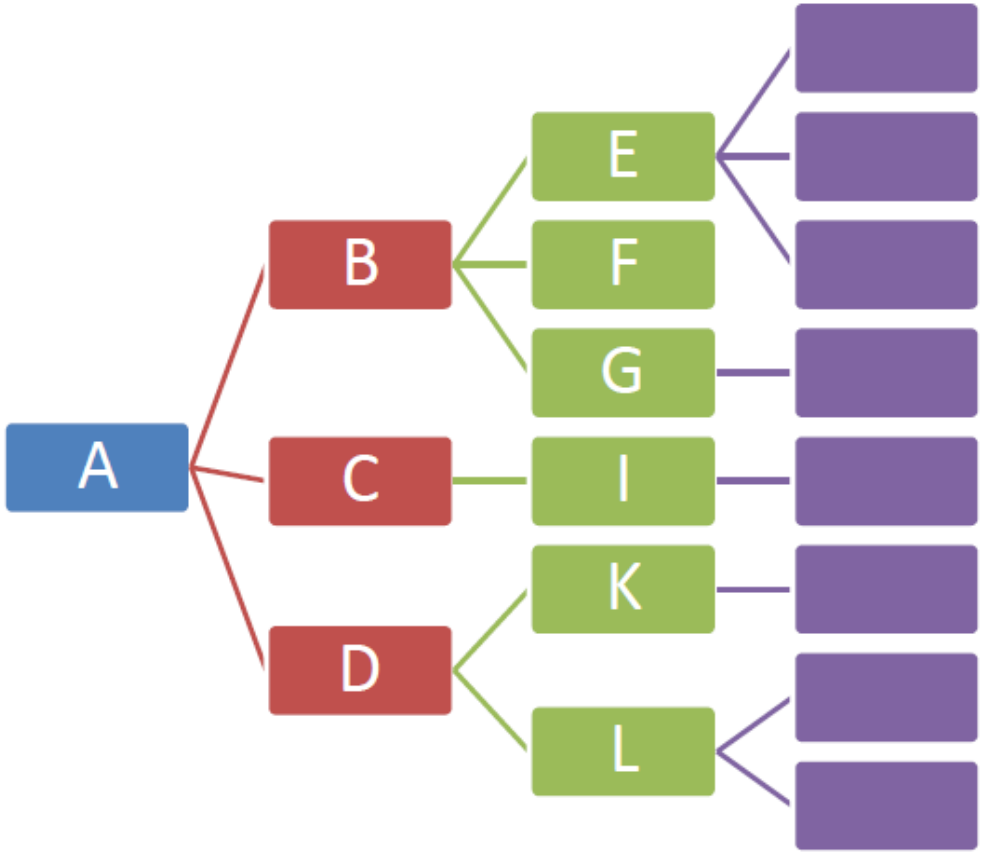
# Ego-centric networks

COM 5506

- A subset of a graph including a particular node (“ego”) and its neighbors (“alters”) (and neighbors of neighbors…)



# Family tree




Is this an undirected or directed graph?

What are the similarities or differences between family tree and ego-network?

# Network density

COM 5506

## Network Density

Potential Connections:

$$PC = \frac{n * (n-1)}{2}$$

Network Density:

$$\frac{\text{Actual Connections}}{\text{Potential Connections}}$$

Examples:

A

Nodes (n): 2
Potential Connections: 1 (2*1/2)
Actual Connections: 1
Network Density: 100% (1/1)

B

Nodes (n): 3
Potential Connections: 3 (3*2/2)
Actual Connections: 3
Network Density: 100% (3/3)

C

Nodes (n): 3
Potential Connections: 3 (3*2/2)
Actual Connections: 2
Network Density: 66.7% (2/3)

# Lab 0 Network representation and data structure

COM 5506

- node vs. edge
- graph vs. map vs. matrix vs. edge list
- attributes vs. relations

- Adjacency matrix
- Edge list
- Adjacency list

# Understanding network data & attribute data structure difference

COM 5506

| ID | Name | Gender | Age | Education |
|---|---|---|---|---|
| 1 | Alice | F | 21 | college |
| 2 | Bob | M | 34 | secondary |
| 3 | John | M | 52 | post-gradu |
| 4 | Matt | M | 12 | primary |
| 5 | Carol | F | 70 | primary |

| | Alice | Bob | John | Matt | Carol |
|---|---|---|---|---|---|
| Alice | 0 | 1 | 0 | 0 | 1 |
| Bob | 1 | 0 | 0 | 1 | 0 |
| John | 0 | 0 | 0 | 0 | 0 |
| Matt | 0 | 1 | 0 | 0 | 1 |
| Carol | 1 | 0 | 0 | 1 | 0 |

- If one additional case is added, the number of size will be increased by?
  - k, k being the constant for the number of columns, in this case 4.

- If one additional case is added, the number of size will be increased by?
  - 2n+1, if you don't count the tag column and row

# Understanding network data & attribute data structure difference

COM 5506

| ID | Name | Gender | Age | Education |
|---|---|---|---|---|
| 1 | Alice | F | 21 | college |
| 2 | Bob | M | 34 | secondary |
| 3 | John | M | 52 | post-gradu |
| 4 | Matt | M | 12 | primary |
| 5 | Carol | F | 70 | primary |

| | Alice | Bob | John | Matt | Carol |
|---|---|---|---|---|---|
| Alice | 0 | 1 | 0 | 0 | 1 |
| Bob | 1 | 0 | 0 | 1 | 0 |
| John | 0 | 0 | 0 | 0 | 0 |
| Matt | 0 | 1 | 0 | 0 | 1 |
| Carol | 1 | 0 | 0 | 1 | 0 |

- Attribute data: rows are cases, columns are variables or features, a particular cell represents a case's value on a particular variable.
- Rectangular data array

- Network data: the first row and column are symmetric, both represent cases or nodes; a particular cell represents relationship between two cases.
- Square array

# Matrix

Matrix: A rectangular array of numbers, e.g., $A \in \mathbb{R}^{m \times n}$:

$$A = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{pmatrix}$$

Vector: A matrix consisting of only one column (default) or one row, e.g., $x \in \mathbb{R}^n$

$$x = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}$$

- Elements of a matrix are identified by their “addresses”
- i=row, j= column
- When i=j, the diagonal

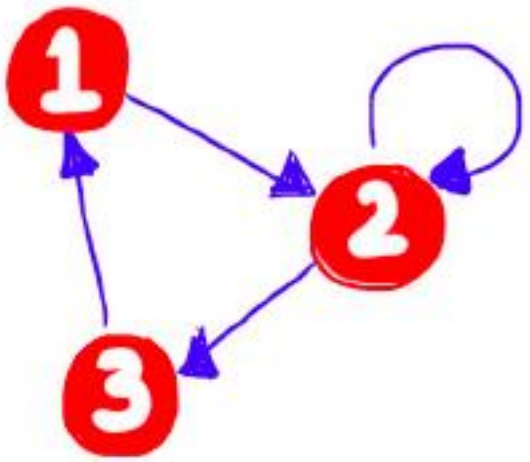
# Which matrix represents the network?

COM 5506

A) $\begin{bmatrix} 0 & 1 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix}$

B) $\begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$

C) $\begin{bmatrix} 0 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix}$



# Matrix permutation

COM 5506

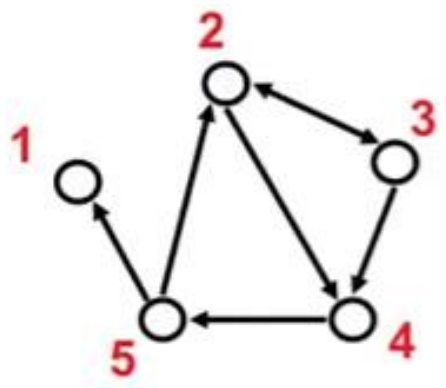
- Permutation – shifting rows and columns while retaining the nature of the objects described in the matrix
- The permutated matrix makes the pattern more obvious

| | Bob | Carol | Ted | Alice |
|---|---|---|---|---|
| Bob | ~ | 1 | 1 | 0 |
| Carol | 0 | ~ | 1 | 0 |
| Ted | 1 | 1 | ~ | 1 |
| Alice | 0 | 0 | 1 | ~ |

| | Bob | Ted | Carol | Alice |
|---|---|---|---|---|
| Bob | ~ | 1 | 1 | 0 |
| Ted | 1 | ~ | 1 | 1 |
| Carol | 0 | 1 | ~ | 0 |
| Alice | 0 | 1 | 0 | ~ |

# Adjacency matrix example

COM 5506



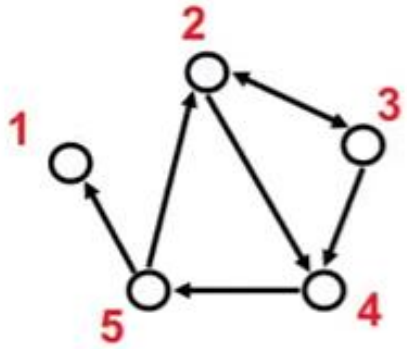

$$A = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 1 & 1 & 0 & 0 & 0 \end{pmatrix}$$

- $A_{ij}$ = 1 if node i has an edge to node j
- $A_{ij}$=0 if node i does not have an edge to j
- $A_{ii}$ = 0 unless the network has self-loops

# Reading degrees from matrix

COM 5506

- Out degree = $\sum_{j=1}^{n} A_{ij}$
- In degree = $\sum_{i=1}^{n} A_{ij}$
- For instance, out degree for node 3 is 2 – the sum of non-zero entries in the 3rd row; the indegree for node 3 is 1 – the sum of non-zero entries in the 3rd column

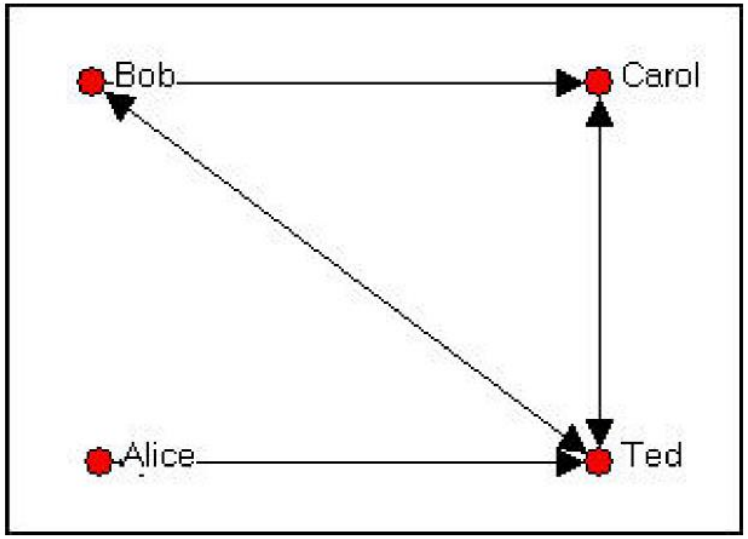



$$A = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 1 & 1 & 0 & 0 & 0 \end{pmatrix}$$

$$\begin{pmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 1 & 1 & 0 & 0 & 0 \end{pmatrix} \qquad \begin{pmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 1 & 1 & 0 & 0 & 0 \end{pmatrix}$$

# Understanding matrix

- Another way of understanding relationship matrix is that it tells us how many paths of **length one** are there from each actor to each other actor.




| | Bob | Carol | Ted | Alice |
|---|---|---|---|---|
| Bob | ~ | 1 | 1 | 0 |
| Carol | 0 | ~ | 1 | 0 |
| Ted | 1 | 1 | ~ | 1 |
| Alice | 0 | 0 | 1 | ~ |

# Linear Algebra review
# Multiplication

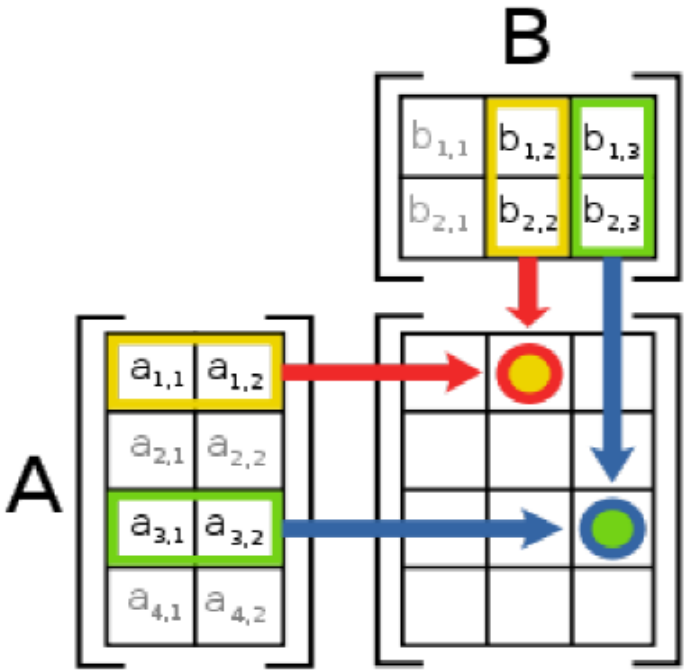
COM 5506

- If $A \in \mathbb{R}^{m \times n}$, $B \in \mathbb{R}^{n \times p}$, then their product $AB \in \mathbb{R}^{m \times p}$
  - Number of columns of $A$ **must** equal number of rows of $B$
- We can compute the product $C = AB$ using this formula:

$$C_{ij} = \sum_{k=1}^{n} A_{ik} B_{kj}$$

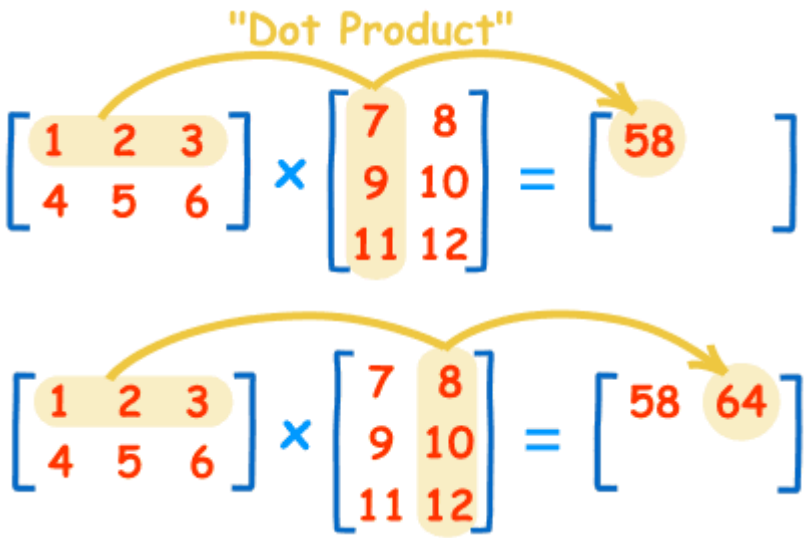
- Associative: $(AB)C = A(BC)$
- Distributive: $A(B + C) = AB + AC$
- Non-commutative: $AB \neq BA$

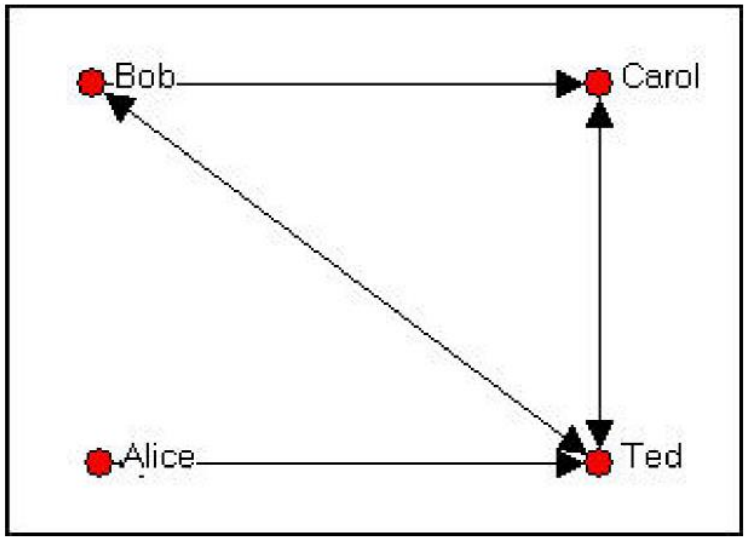
$$(AB)(x) = A(B(x))$$



# Matrix multiplication

COM 5506



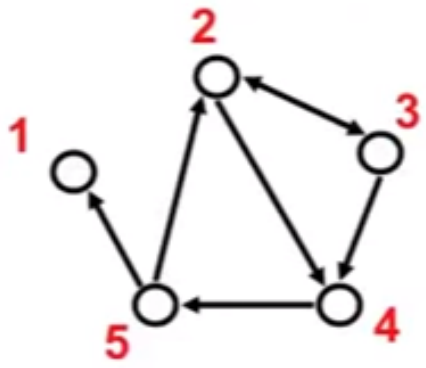

$$\begin{array}{c} \\ B \\ C \\ T \\ A \end{array} \begin{array}{c} \begin{array}{cccc} B & C & T & A \end{array} \\ \begin{bmatrix} 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix} \end{array} \times \begin{bmatrix} 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 0 & 1 & 3 & 0 \\ 1 & 1 & 0 & 1 \end{bmatrix}$$

- The square of matrix denotes how many pathways ***of length two*** are there from each actor to each other actor.

# Edge list

Edge list

- 2, 3
- 2, 4
- 3, 2
- 3, 4
- 4, 5
- 5, 2
- 5, 1




- Edge list is easier to work if the network is
  - Large
  - Sparse

# Adjacency list

- Quickly retrieve all neighbors for a node

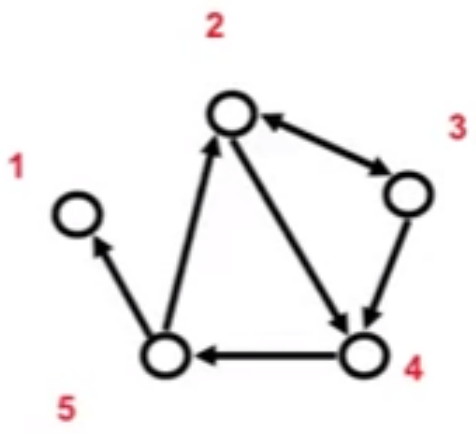
  1:

  2: 3 4

  3: 2 4

  4: 5

  5: 1 2



# From Graph to Matrix

- Using excel, transform the 13-node Arpanet graph to a matrix: “1 for connected” and 0 for “not connected”

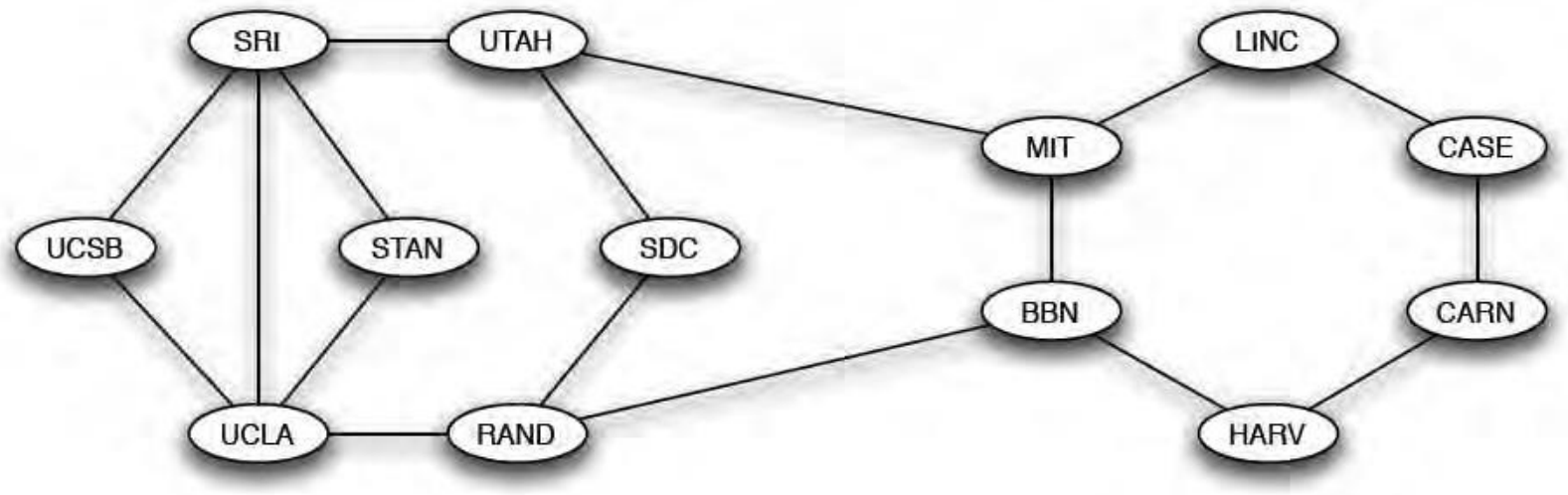



Figure 2.3: An alternate drawing of the 13-node Internet graph from December 1970.

# Matrix Data of Arpanet

COM 5506

| | UCSB | SRI | STAN | UCLA | UTAH | SDC | RAND | MIT | BBN | LINC | CASE | CARN | HARV |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| UCSB | | | | | | | | | | | | | |
| USRI | | | | | | | | | | | | | |
| STAN | | | | | | | | | | | | | |
| UCLA | | | | | | | | | | | | | |
| UTAH | | | | | | | | | | | | | |
| SDC | | | | | | | | | | | | | |
| RAND | | | | | | | | | | | | | |
| MIT | | | | | | | | | | | | | |
| BBN | | | | | | | | | | | | | |
| LINC | | | | | | | | | | | | | |
| CASE | | | | | | | | | | | | | |
| CARN | | | | | | | | | | | | | |
| HARV | | | | | | | | | | | | | |

For an “unweighted” graph like the current one, enter “1” in a cell for an existing edge and “0” otherwise. For an “undirected” graph like the current one, enter values to cells of one (rather than both) off-diagonal.

# Edge list data of Arpanet

COM 5506

| Node1 | Node2 | Weight |
|---|---|---|
| UCSB | SRI | |
| UCSB | UCLA | |
| … | … | |
| … | … | |
| … | … | |
| … | … | |
| … | … | |
| … | … | |
| … | … | |
| … | … | |
| … | … | |
| … | … | |
| HARV | CARN | |

- An edge list has two required columns to record the ID of each pair of nodes, and an optional column, weight, to record the strength of each edge.
- Edge list is more efficient to represent sparse and large networks.

# Next week

COM 5506

- Node Centrality and power
- Python intro